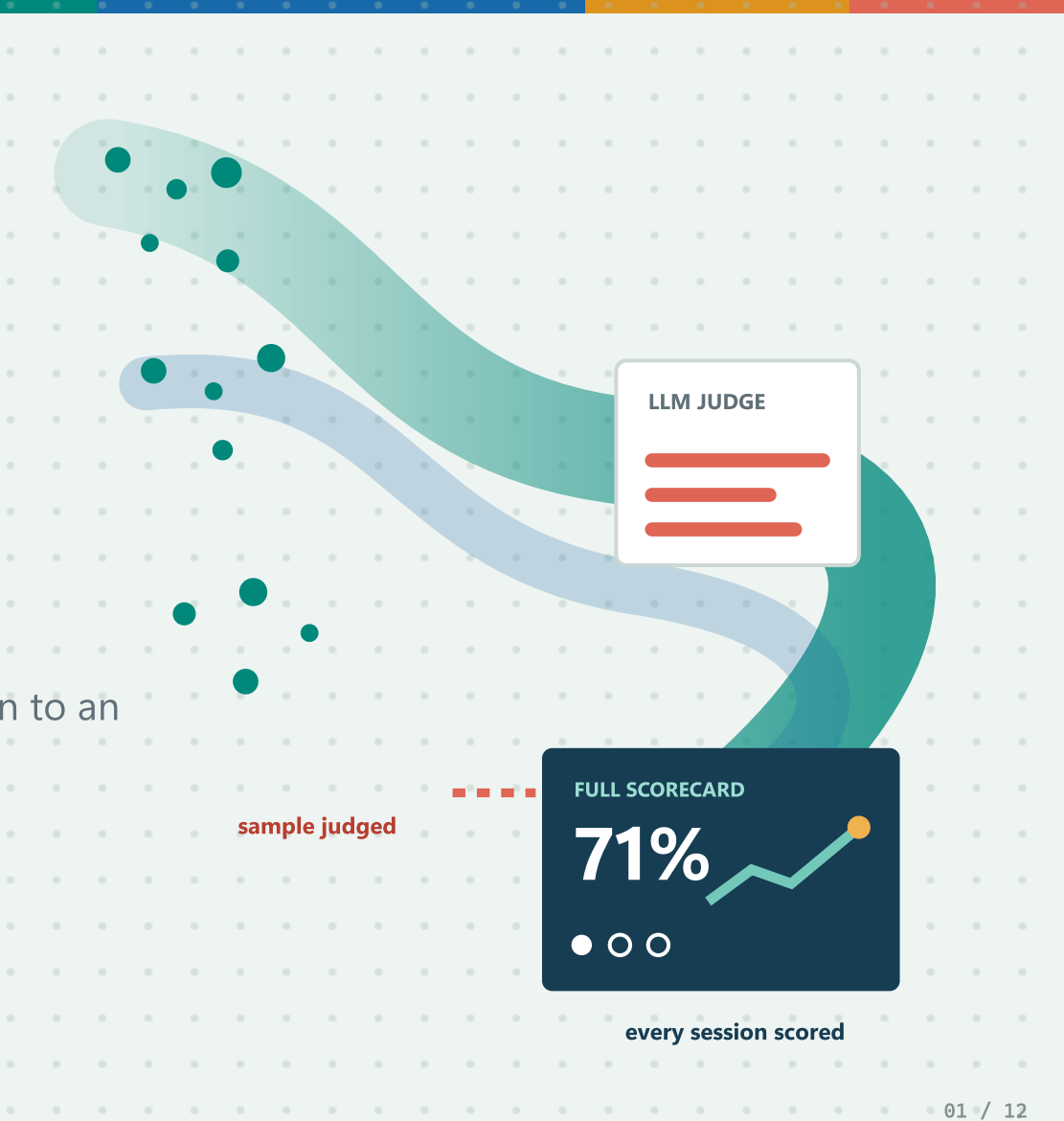


AGENT 365 EVALUATIONS

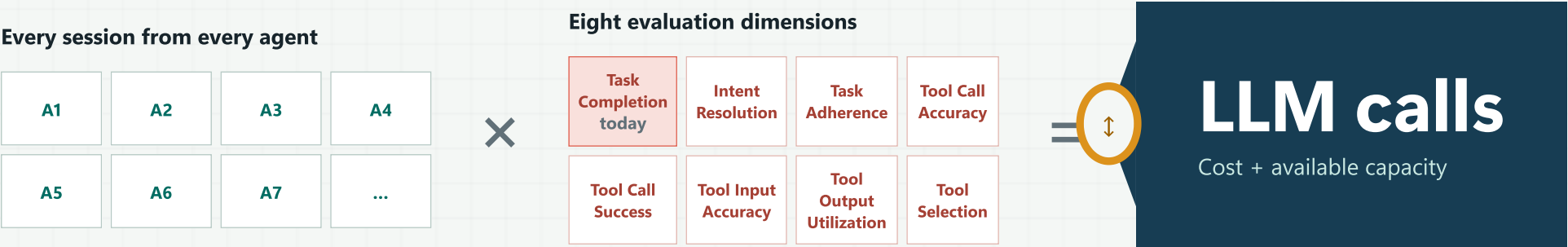
Sampling for Agent 365 Evaluations

Evaluate every agent without sending every session to an LLM judge.

SAMPLING RESEARCH · V4 + V6



The ambition multiplies faster than capacity.

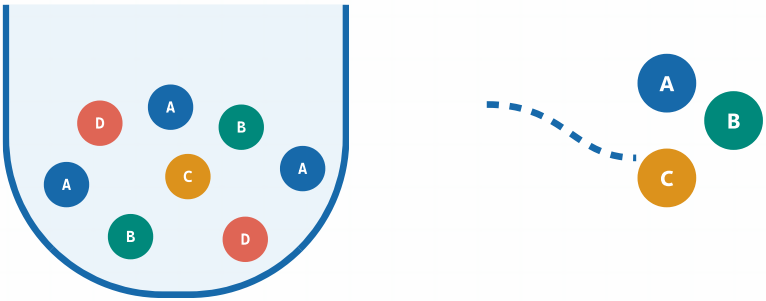


Sampling turns an unbounded evaluation bill into a controlled capacity decision.

Two simple ways to choose what gets judged.

Random Sampling

Every session gets an equal chance.



Simple

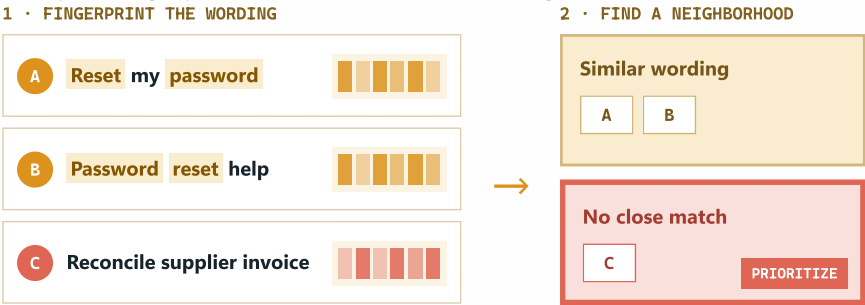
Low operational complexity

Defensible

Preserves equal chance

MinHash LSH

Compact fingerprints reveal unfamiliar wording.



Fast

Avoids all-pairs comparison

Lexical

Words, phrases, tool patterns

Random represents fairly. MinHash searches for lexical novelty.

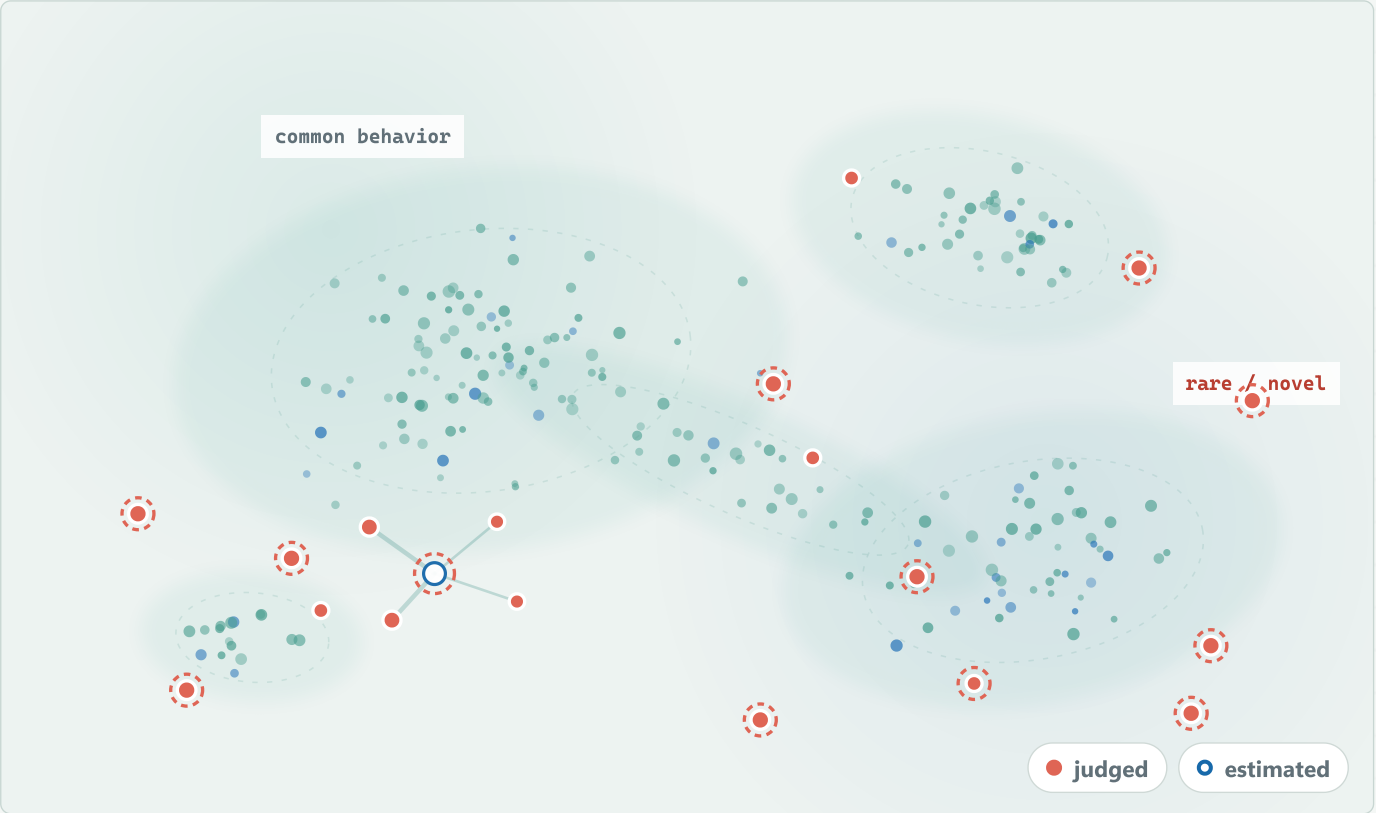
Map meaning, sample broadly, estimate locally.

1 Map
A whole session becomes one point based on meaning.

2 Sample
Judge points that broaden behavioral coverage.

3 Estimate
Nearby judged points inform unjudged points.

● Embedding = map ○ IDW = estimate



We tested against a known answer.



Success means accuracy and breadth.

Outcome error (MAE)

How close is the sampled pass rate to the full-population pass rate?



$|69\% - 72\%| = 3 \text{ points}$ · lower is better

Concept coverage

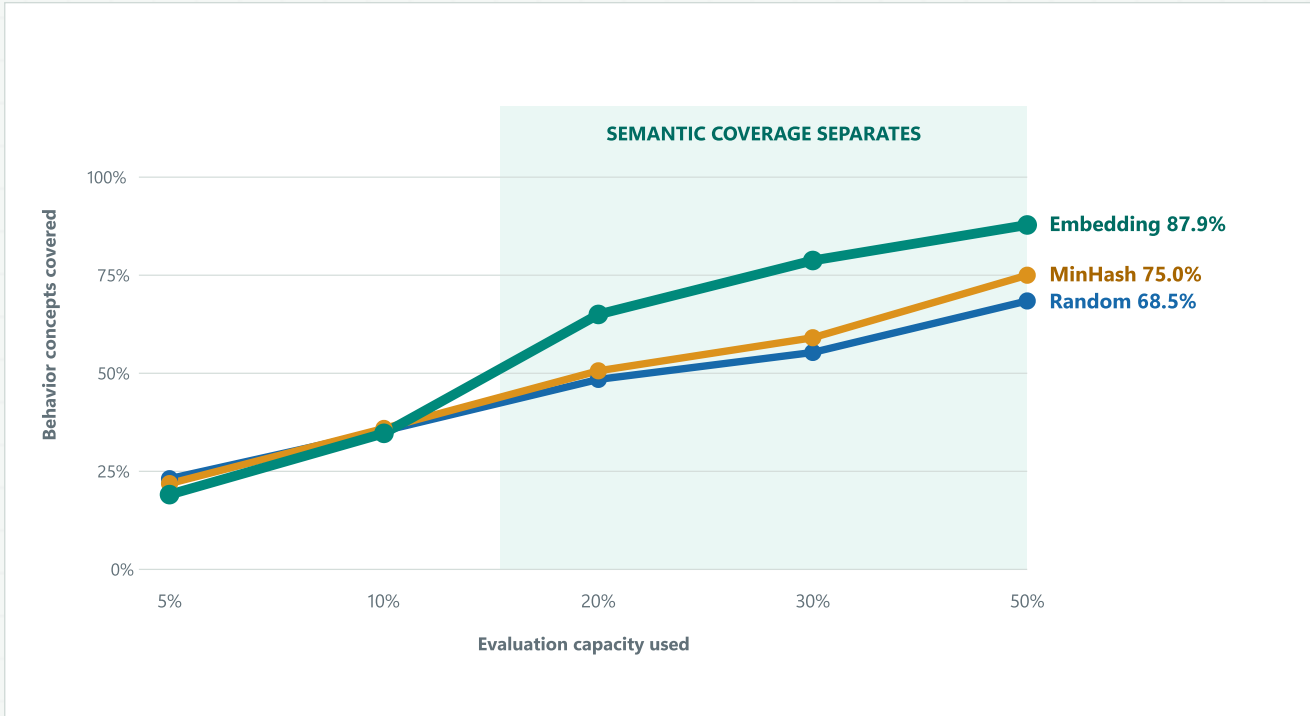
How much of the population's behavioral variety did the sample touch?



$120 \div 480 = 25\%$ · higher is better

Same sessions · same whole-session token budgets · repeated deterministic replays

Semantics won coverage; simplicity won the first release.



+19.4 pts

more concept coverage than Random at 50% capacity

Coverage is not the whole story

Random had lower headline pass-rate error at **4 of 5** budgets.

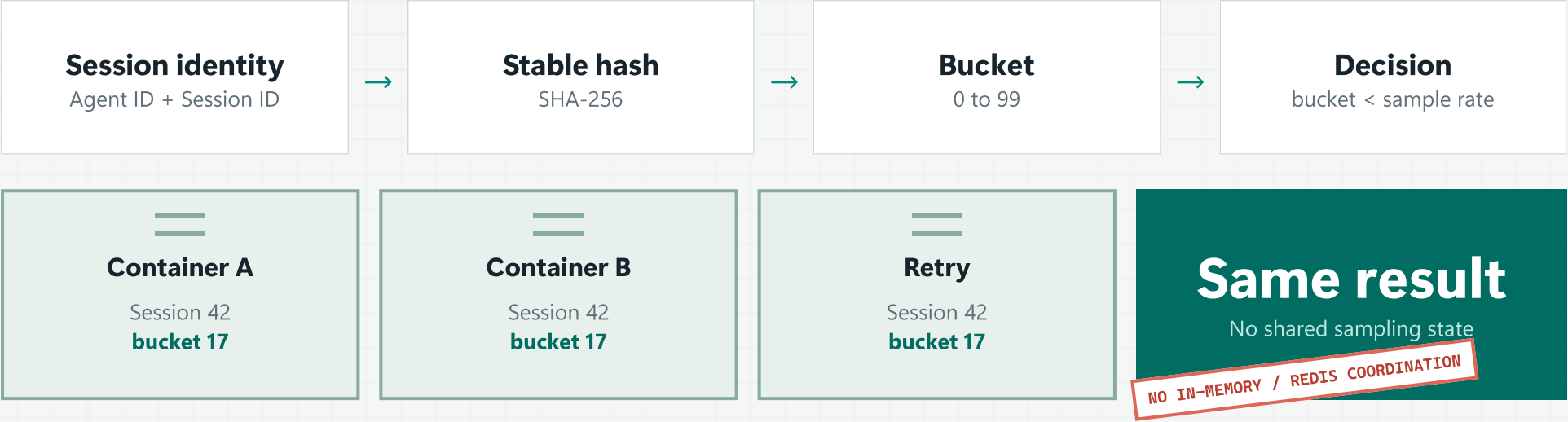
NOW

Deterministic
Random

NEXT

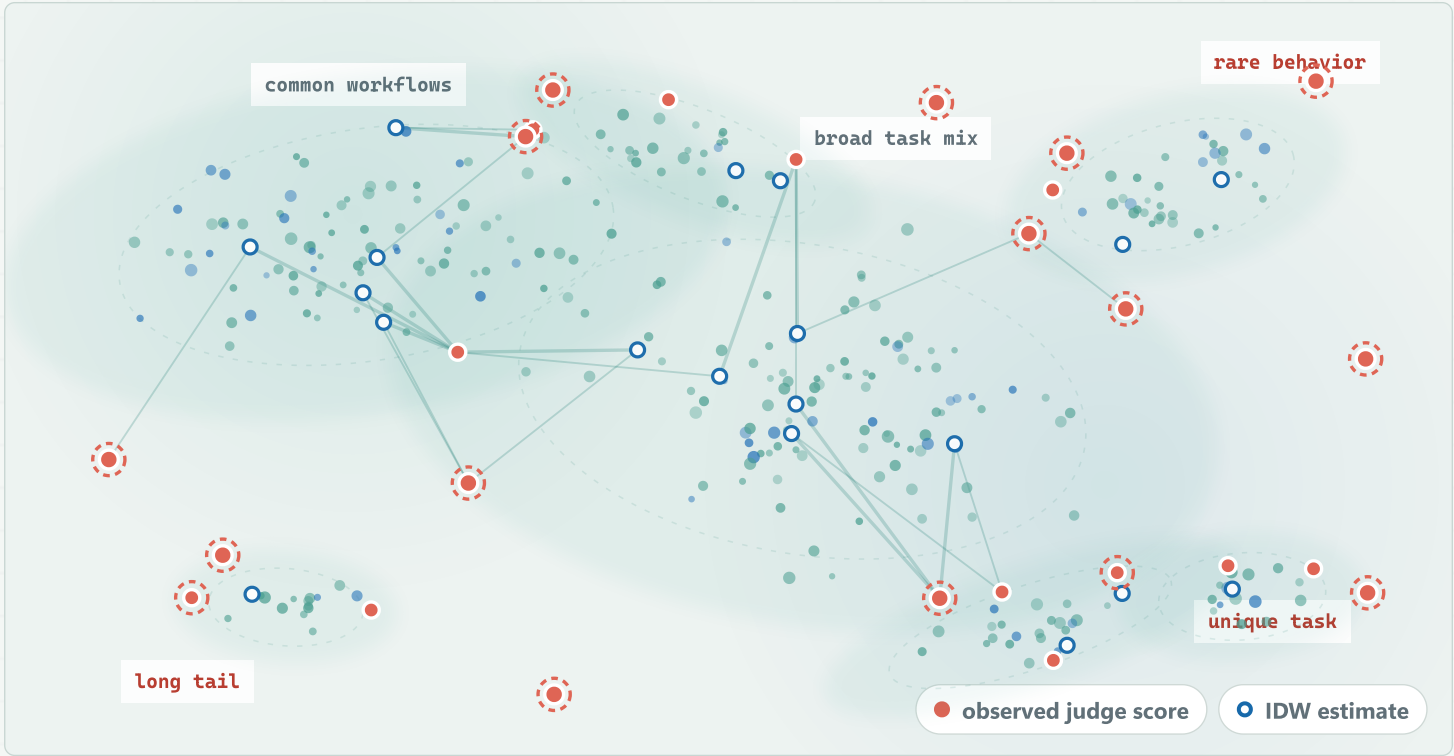
→ Embedding +
IDW

The same session gets the same answer everywhere.



Same session + same rate = same decision in every container and retry. Then apply per-agent minimums and run/capacity limits.

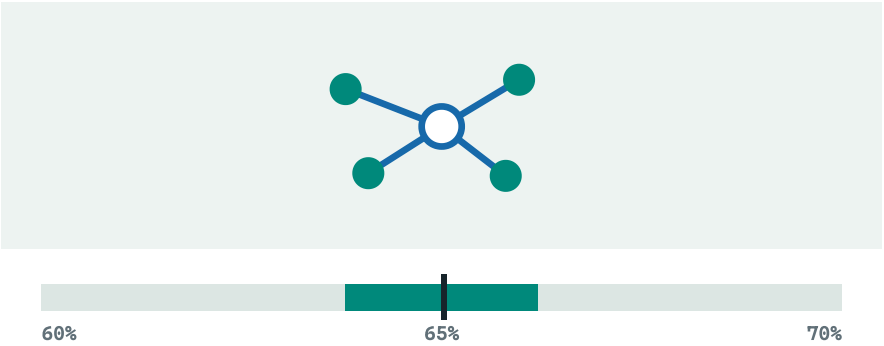
Judge the evidence-rich sample; score the full picture.



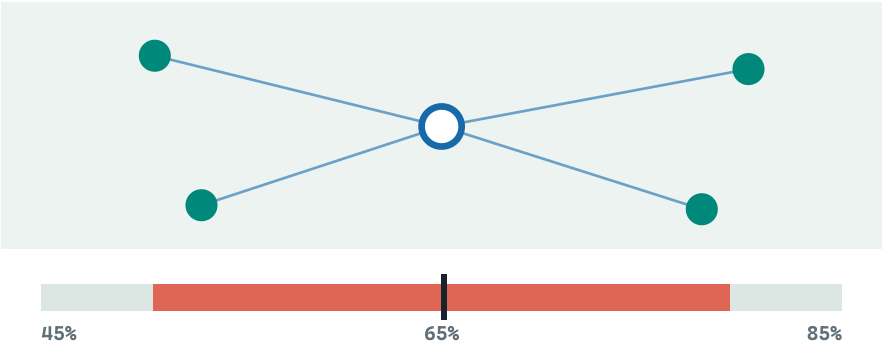
- Find**
Novel and rare sessions
- Reveal**
Long-tail tasks inside super agents
- Judge**
Only the evidence-rich sample
- Estimate**
Every eligible unjudged session
- Broader discovery + population-wide estimated scores**

Confidence depends on how far we reached.

Nearby evidence → narrower range



Distant evidence → wider range



$$\text{Estimated score} \pm (\text{behavior sensitivity} \times \text{weighted neighbor distance})$$

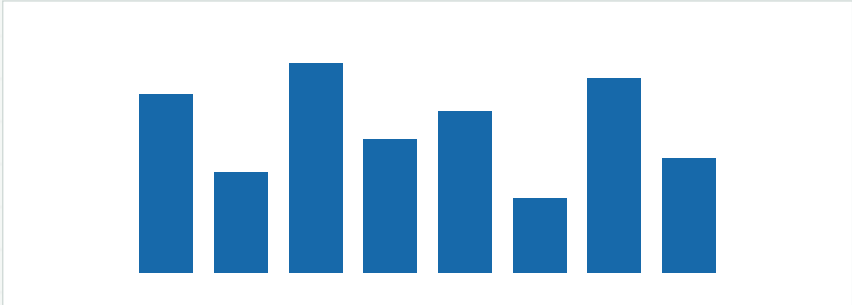
Example: 65% estimated pass probability → 55%–75% conditional range

Conditional geometric sensitivity · not a statistical confidence interval

How small can the representation become?



Native embedding · 1,536 values per session



Reduced representation · 8 values per session

PCA	SVD	GRP	Random coordinate mean
2,500 sessions	30 replay seeds	5 traffic patterns	5 judge budgets

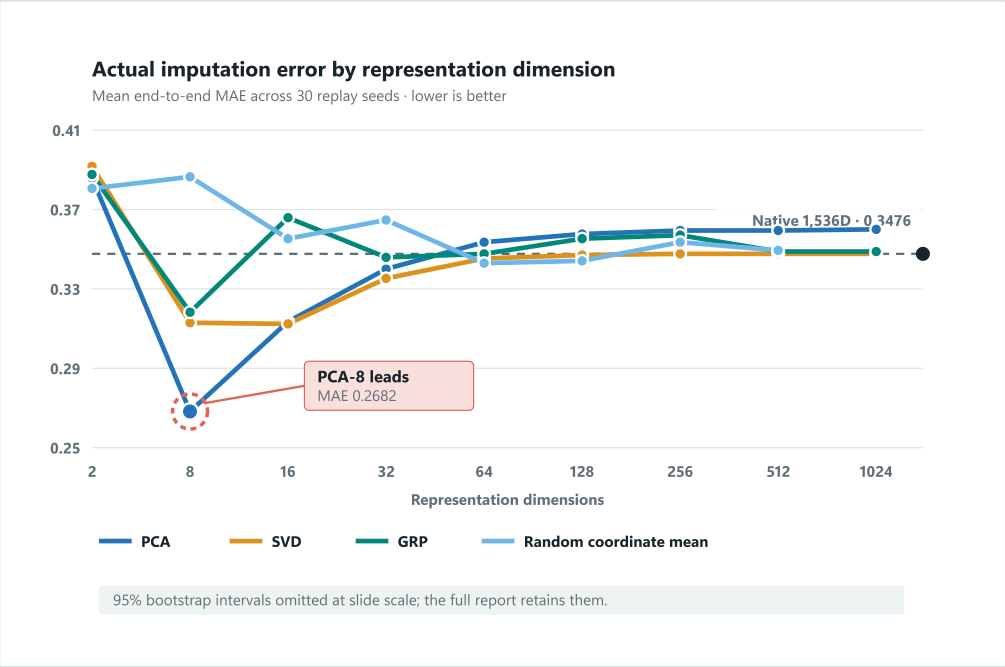
reduce → select → impute with IDW → compare with expected labels

Eight dimensions can unlock the semantic path.

8D **Small representation**
from 1,536 native dimensions

23% **Lower end-to-end MAE**
0.2682 vs 0.3476 native

119× **Smaller stored representation**
0.129 MB vs 15.36 MB



Proposed unlock: store eight components in ESP and compute similarity directly.

Synthetic-label research result. Confirm reducer choice, ESP schema/compliance, and production-scale performance.